OUZZANE Réda
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EDUCATION

ENS Paris-Saclay — **Gif-sur-Yvette, France**
Mathématiques, Vision & Apprentissage (MVA) - Research Master's (M2) — September 2026 - September 2027
- **Relevant Coursework:** Stochastic Calculus & Generative Modeling, Optimal Transport, Foundations of Large Language Models, Computational Statistics, Convex Optimization, Machine Learning for Time Series

École Centrale Lille — **Lille, France**
French "Grande École" Engineering Program - GPA: 3.78/4 — September 2023 - September 2026
- **Relevant Coursework:** Machine Learning, Deep Learning, Probability & Statistics, Markov Chains, Advanced Statistical Estimation, Optimization, Computer Vision, Image Modeling and Denoising

Politecnico di Torino — **Turin, Italy**
Exchange Semester in Data Science and Engineering - GPA: 3.57/4 — February 2025 - July 2025

Lycée Charlemagne — **Paris, France**
Preparatory Classes for French Engineering Schools (PCSI / PC)* — September 2020 - July 2023

RESEARCH & PROFESSIONAL EXPERIENCE

CNRS - CRIStAL Lab — **Lille, France**
Research Project – Diffusion Models & Numerical Inversion — September 2025 - March 2026
- Investigated the numerical reversibility of latent diffusion through its ODE formulation under the supervision of Patrick Bas, developing a Stable Diffusion 2.1 generation–inversion pipeline.
- Analyzed noise-reconstruction error across 4,806 runs with six prompts and up to four seeds, examining sensitivity to guidance mismatch, solver settings, initial noise variance and VAE round-trips.

KPMG — **Paris, France**
Data Science Intern - NLP & Document Intelligence — April 2026 - September 2026
- Developed a Python document understanding pipeline combining Docling parsing, semantic retrieval with multilingual Qwen3 embeddings and French/English KPI terminology, rule-based extraction.
- Evaluated extraction across eight insurers and two fiscal years, achieving 95.9% coverage (445/464 KPI values), with source provenance and consistency checks to support error analysis.
- Designed comparative experiments with three LLMs across ten financial reporting tasks, generating 30 reports and tracking prompts, outputs, inference latency and token usage.

ACADEMIC PROJECTS

Independent Research Project — **Paris, France**
Reward-Guided Sampling for Diffusion Models — October 2026
- Implemented three test-time guidance methods (full-gradient Tweedie, isotropic covariance approximation, LiDAR) for Stable Diffusion 1.5 using PyTorch, differentiable rewards and DDIM sampling.
- Benchmarked 128 images across eight prompts and four seeds on NVIDIA H100; full-gradient Tweedie guidance improved luminance reward in 32/32 paired comparisons, analyzing CLIPScore, HPSv2 and computational trade-offs.

Politecnico di Torino — **Turin, Italy**
Sparse Fine-Tuning of Vision Transformers in Federated Learning — February 2025 - July 2025
- Implemented a federated learning pipeline for a DINO-pretrained ViT-S/16 on CIFAR-100, comparing centralized training and FedAvg under IID and non-IID client distributions.
- Benchmarked five gradient-masking strategies based on Fisher information, weight magnitude and random selection, analyzing how sparsity and communication frequency affect convergence and test accuracy.

Lille University Hospital — **Lille, France**
Constrained Optimization for Medical Staff Scheduling — September 2024 - February 2025
- Developed a Python tool for constrained medical staff scheduling using metaheuristic optimization, reducing the objective value by ~9% versus the hospital's reference schedule on the evaluated case.

SKILLS & INTERESTS

Technical Skills: Python, C, SQL, PyTorch, scikit-learn, Hugging Face Diffusers, NumPy, pandas, LaTeX
Languages: French (Native), English (Fluent), Spanish (Intermediate), Italian (Beginner)
Interests: Team sports (Football, Basketball), Travel, Literature